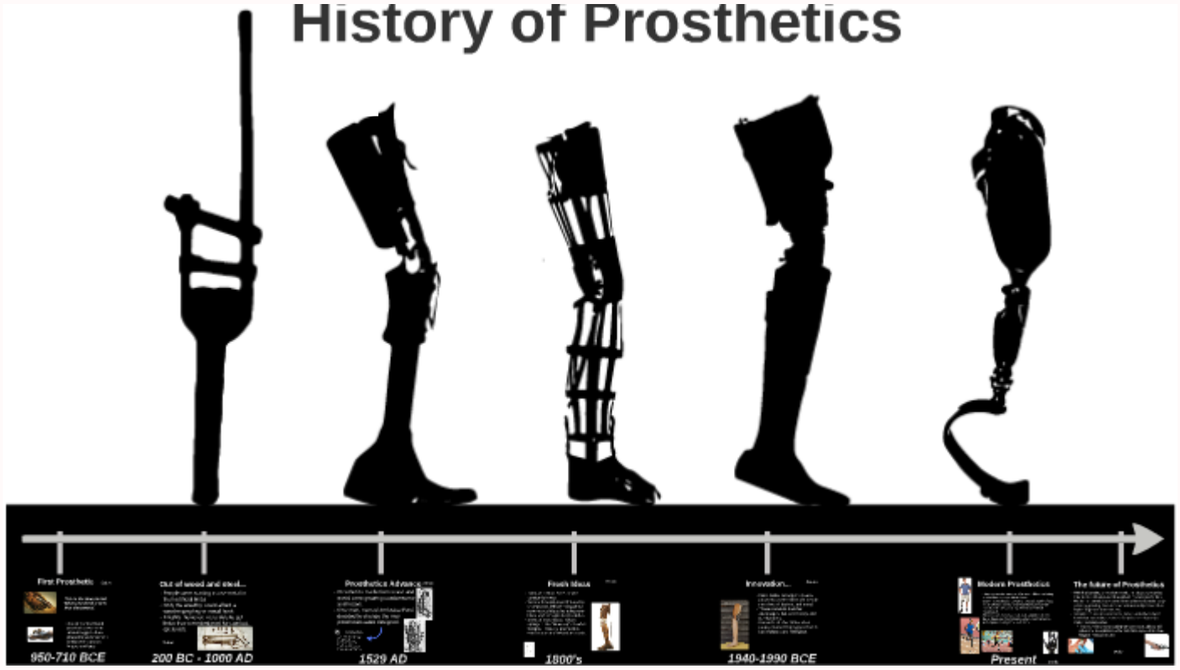
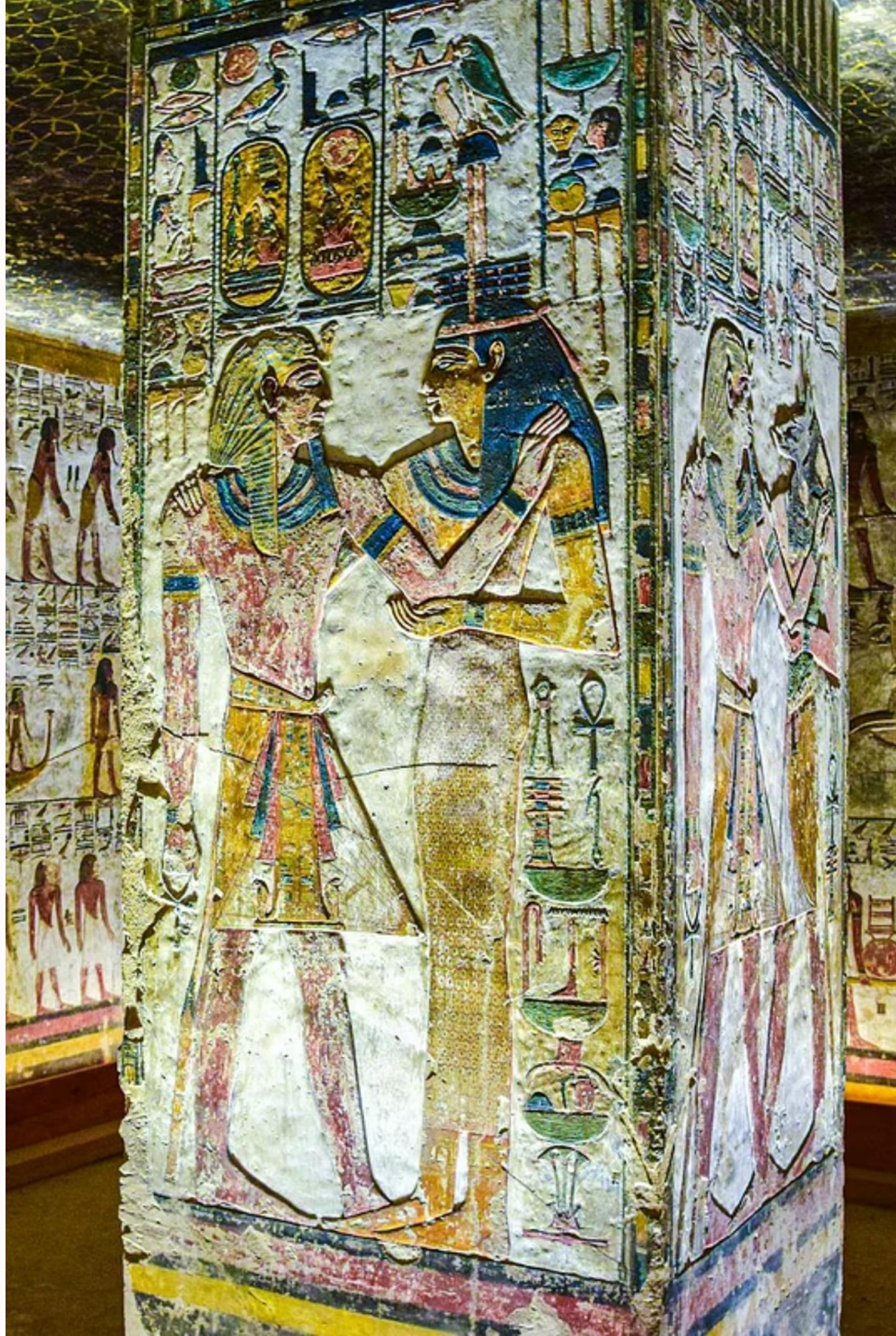
Prosthetic Limbs: From Ancient Ancient Solutions to Neuro-Controlled Bionics

AYUSH KUMAR (RA2511013010078)
ANUJ YADAV (RA2511060010026)
VARSHITH REEDY (RA2511060010027)
AARIZ NUMAN AHMED (RA2511060010028)



INTRODUCTION:

The topic “Prosthetic Limbs: From Ancient Solutions to Neuro-Controlled Bionics” Explores the remarkable evolution of prosthetic technology, tracing its journey from early, rudimentary devices to today's advanced neuro-controlled bionics. This presentation delves into how human ingenuity has continuously sought to restore function and enhance the lives of individuals throughout history.



Prehistoric & Ancient Innovations: The Dawn of Prosthetics

The concept of replacing lost limbs dates back millennia. Early evidence suggests basic solutions for both functional and aesthetic purposes.

Functional Necessity

Simple peg legs and hand hooks provided mobility and the ability to perform daily tasks, crucial for survival.

Early Materials

Wood, leather, and rudimentary metals were the primary materials, shaped with basic tools to create prosthetic devices.

Medieval to Renaissance: War & Craftsmanship

The medieval period, marked by frequent conflicts, spurred significant advancements in prosthetic design, moving towards more articulated and weapon-integrated solutions.

Armored Prosthetics

Knights and soldiers received custom-made iron hands or gauntlets, often designed to hold shields or weapons, blurring the line between limb replacement and battle gear.

Articulated Designs

Pioneers like Ambroise Paré introduced more sophisticated, articulated devices with springs and ratchets, offering improved grip and manipulation capabilities.

Early Mechanization

The Renaissance saw further refinement, integrating basic mechanical principles for enhanced function, albeit still heavy and often uncomfortable.

The Industrial Revolution: Mass Production & Material Science

The 19th and early 20th centuries brought an explosion of innovation, driven by new manufacturing techniques and scientific understanding of materials.

01

Lightweight Materials

The introduction of lighter metals and vulcanized rubber significantly improved comfort and wearability for amputees.

02

Standardized Components

Mass production allowed for more affordable and widely available prosthetic components, moving away from purely bespoke designs.

03

Improved Sockets

Better understanding of anatomy led to more precisely fitted sockets, reducing chafing and increasing stability, a critical factor for usability.

Modern Era: The Rise of Bionics and Integration

Post-WWII, advancements accelerated, leading to prosthetics that not only mimic appearance but also integrate with the body's nervous system.

Neural Interfaces

Direct nerve connection allows for intuitive control, translating thought into movement with unprecedented precision.

Sensory Feedback

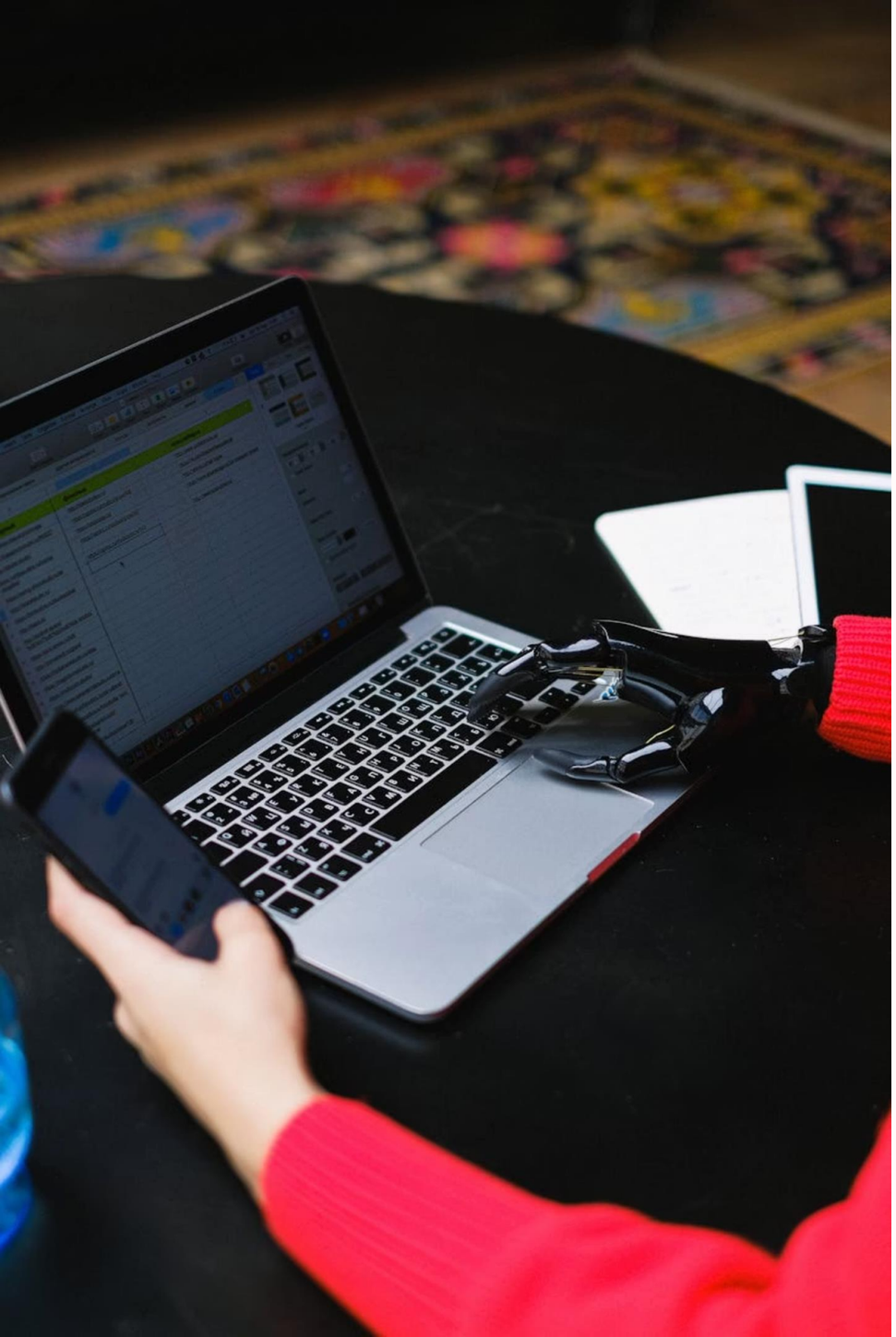
Emerging technologies aim to restore tactile sensation, allowing users to "feel" what their prosthetic touches, further enhancing integration.

Advanced Robotics

Miniature motors and sensors enable lifelike dexterity and responsive feedback, improving grasping and object manipulation.

Custom Fabrication

3D printing and CAD software create highly customized, lightweight, and perfectly fitted prosthetics for each individual.



The Future of Prosthetics: Beyond Restoration

The journey from simple tools to complex bionic limbs reflects humanity's continuous drive for improvement. The future promises prosthetics that not only restore lost function but also potentially enhance human capabilities, offering new frontiers in biomedical engineering.

BIOMEDICAL ENGINEERING

NEUROPROSTHETICS

ASSISTIVE TECHNOLOGY

THANK YOU